

Shehbaz Tariq

Postdoctoral Researcher at SnT¹
Kyung Hee University, South Korea

Interdisciplinary Centre for
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Education

- **Kyung Hee University, Yongin-si, Korea** Mar. 2021 – Aug. 2025
Ph.D. in Electronic and Information Convergence Engineering
Specialization in Quantum Computing and Machine Learning (MS leading to Ph.D)
CGPA: 4.28 / 4.50
- **University of Engineering and Technology, Peshawar, Pakistan,** Aug. 2016 – Sep. 2020
Bachelor of Engineering in Electrical Engineering
CGPA: 3.87 / 4.00
Final Year Project: AIOT-based monitoring app for smart cities

Research Interests

General research interest is quantum information processing for communication systems, emphasizing integrating quantum technologies with 6G and future networks, quantum network design and applications, and utilizing large AI models to develop integrated hybrid algorithms for next-generation networks.

Teaching Interests

Quantum Computing, Quantum Communications, Large AI applications for Quantum Information Processing, Machine Learning Programming, and Linear Algebra.

Experience

Work Experience

- **SnT, University of Luxembourg, Kirchberg, Luxembourg** Apr. 2025 – Present
Postdoctoral Researcher at Signal Processing and Communications (SIGCOM) Research Group
Conduct research on Quantum MIMO systems and Quantum Reservoir Computing.
Develop optimization and quantum machine learning techniques for wireless communication systems.
Collaborate with academic and industrial partners on large-scale European research projects.
- **Kyung Hee University, Yongin-si, Korea** Mar. 2021 – Mar. 2024
CQILAB Fellow & Research Assistant
Department of Electronics and Information Convergence Engineering
Mentored undergraduate and graduate students in their research in quantum information sciences.
Lead the group seminars on Quantum Machine Learning and AI.
Managed and coordinated different teams working on large-scale R&D projects including national research foundation (NRF)-funded projects.
- **Queen's University Belfast, Northern Ireland, UK** Oct. 2023
Visiting Scholar
School of Electronics, Electrical Engineering and Computer Science
Organize seminars on Large AI applications for quantum computing, 6G and beyond.
Managed and coordinated different teams working on quantum optimization for 6G applications.

¹Interdisciplinary Centre for Security, Reliability and Trust (SnT)

- **University of Engineering and Technology, Peshawar, Pakistan**
Research Intern
 National Center of Artificial Intelligence (NCAI)
 Learned the core concepts of Machine Learning and AI.
 Assisted the Smart Flood and Safe City Teams.

Mar. 2019 – Aug. 2019

Publications & Presentations

Under Review/Preprints

- [1] S. Tariq, M. Talha, S. Chatzinotas, and H. Shin, “Towards Quantum Enhanced Adversarial Robustness with Rydberg Reservoir Learning” under review, 2026.
- [2] S. Tariq, J. ur Rehman, and S. Chatzinotas, “Quantum MIMO Diversity over Discrete-Variable Crosstalk Channels”, under review, 2026.
- [3] J. Kaldari, S. Tariq, S. Al-Kuwari, S. Y.-C. Chen, S. Chatzinotas, and H. Shin, “Quantum Reinforcement Learning: Recent Advances and Future Directions” under review, 2026.
- [4] L. Oleynik, S. Tariq, and S. Chatzinotas, “Optimal GHZ-State Distribution in LOSR Quantum Networks via Local Decoding from Information Sets” under review, 2026.

Journal Papers

- [1] S. Tariq and S. Chatzinotas, “Design and Optimization of Adaptive Diversity Schemes in Quantum MIMO Channels” *npj Quantum Information*, Jun. 2026.
- [2] N. Lae, S. Tariq, S. N. Paing, J. W. Setiawan, T. Q. Duong, and H. Shin, “Counterfactual Quantum Secret Sharing” *IEEE Transactions on Communications*, Jun. 2025.
- [3] B. E. Arfeto, S. Tariq, U. Khalid, T. Q. Duong, and H. Shin, “GenSC-6G: A Prototype Testbed for Integrated Generative AI, Quantum, and Semantic Communication” accepted for publication in *IEEE Communications Magazine*, May. 2025.
- [4] S. Tariq, B. E. Arfeto, U. Khalid, S. Kim, T. Q. Duong, and H. Shin, “Deep Quantum-Transformer Networks for Multi-Modal Beam Prediction in ISAC Systems” *IEEE Internet of Things Journal*, 18:29387–29401, Sep. 2024.
- [5] S. Tariq, A. Farooq, J. ur Rehman, T. Q. Duong, and H. Shin, “Efficient Quantum State Estimation with Low-rank Matrix Completion”, *EPJ Quantum Technology*, 11:50, Aug. 2024.
- [6] S. Tariq, U. Khalid, B. E. Arfeto, T. Q. Duong, and H. Shin, “Integrating Sustainable Big AI: Quantum Anonymous Semantic Broadcast”, *IEEE Wireless Communications*, 31:86–99, Jun. 2024.
- [7] S. N. Paing, J. W. Setiawan, S. Tariq, M. T. Rahim, K. Lee, and H. Shin, “Counterfactual Anonymous Quantum Teleportation in the Presence of Adversarial Attacks and Channel Noise”, *Sensors*, 22:7587, Oct. 2022.

Refereed Conference Proceedings

- [1] S. Tariq, J. ur Rehman, and S. Chatzinotas, “Quantum MIMO Diversity Framework Using Approximate Cloning and Probabilistic Purification” in *2026 International Conference on Quantum Communications, Networking, and Computing (QCNC)*, Apr. 2026.
- [2] B. E. Arfeto, S. Tariq, and H. Shin, “Edge Diffusion Model for Enhanced Resolution in Semantic Communication” in *Proc. Korea Information and Communications Society (KICS) Fall Conference*, Gyeongju, Korea, Nov. 2024.
- [3] M. Talha, S. Tariq, and H. Shin, “Recent Advances in Quantum Reservoir Computing” in *Proc. Korea Information and Communications Society (KICS) Fall Conference*, Gyeongju, Korea, Nov. 2024.

- [4] P. M. Min, B. E. Arfeto, S. Tariq, and H. Shin, "Quantum-Augmented LSTM Networks for Stock Price Forecasting" in *Proc. Korea Information and Communications Society (KICS) Fall Conference*, Gyeongju, Korea, Nov. 2024.
- [5] S. Tariq, B. E. Arfeto, H. Jung, and H. Shin, "Beam Prediction in mmWave V2I Communications with Quantum Transformer Networks," in *Proc. International Conference on Information and Communication Technology Convergence (ICTC)*, Jeju, Korea, Oct. 2024
- [6] S. Tariq, J. U. Roh, and H. Shin, "Quantum Reinforcement Learning for Digital Twin Placement in 6G Networks" in *Proc. Joint Conference on Communications and Information (JCCI)*, Busan, Korea, Apr. 2024.
- [7] S. Tariq, M. S. Ulum, A. W. Shaffar, W. Park, S. Kim, and H. Shin, "Multi-Agent Quantum Reinforcement Learning for Digital Twin Placement in 6G Multi-Tier Systems" in *Proc. EAI International Conference on Industrial Networks and Intelligent Systems (EAI INISCOM)*, Danang, Vietnam, Feb. 2024.
- [8] S. Tariq, A. U. Haque, H. Jung, and H. Shin, "Quantum Approximate Optimization for Optimal EV Charging Schedules" in *Proc. Joint Conference on Communications and Information (JCCI)*, Yeosu, Korea, Apr. 2023.
- [9] U. I. Paracha, S. Tariq, M. Choi, and H. Shin, "Bit-Flip Error Mitigation Using Deep Learning," in *Proc. Korea Information and Communication Society (KICS) Winter Conference*, Pyeongchang, Korea, Feb. 2023.
- [10] A. U. Haque, S. Tariq, H. Jung, and H. Shin, "Prototyping a Quantum Clustering Algorithm for NISQ Computing" in *Proc. Korea Information and Communication Society (KICS) Winter Conference*, Pyeongchang, Korea, Feb. 2023.
- [11] S. Tariq, K. Lee, and H. Shin, "Deep Learning Assisted Quantum State Tomography" in *Proc. Korea Information and Communications Society (KICS) Fall Conference*, Gyeongju, Korea, Nov. 2022.
- [12] A. W. Shaffar, S. Tariq, and H. Shin, "Quantum Convolutional Neural Networks for Binary Classification" in *Proc. Korea Information and Communications Society (KICS) Fall Conference*, Gyeongju, Korea, Nov. 2022.
- [13] S. Tariq, A. Farooq, and H. Shin, "Robust Quantum State Tomography with Linear Regression" in *Proc. Korea Information and Communication Society (KICS) Summer Conference*, Jeju, Korea, Jun. 2022.
- [14] A. W. Shaffar, M. A. Ullah, S. Tariq, and H. Shin, "Experimental Benchmarking Between IBM Quantum Computing Systems" in *Proc. Korea Information and Communication Society (KICS) Summer Conference*, Jeju, Korea, Jun. 2022.
- [15] S. Tariq, A. Farooq, J. U. Rehman, and H. Shin, "Error Analysis of Gate and Pulse Model Quantum Circuits" in *Proc. Korea Information and Communication Society (KICS) Winter Conference*, Pyeongchang, Korea, Feb. 2022.

Theses

1. S. Tariq, 'Hybrid Quantum-Classical Computing for 6G Networks: Algorithms, Applications, and System Integration,' Ph.D. dissertation, Department of Electronics and Information Convergence Engineering, Kyung Hee University, Yongin-si, Korea, Feb. 2025, Thesis Advisor: Professor Hyundong Shin.
2. S. Tariq, "AIOT-based monitoring app for smart cities," Bachelor's thesis, Department of Electrical Engineering, University of Engineering and Technology, Peshawar, Pakistan, Aug. 2020, Thesis Advisor: Dr. Muhammad Amir.

Certifications

- Qiskit Global Summer School (2024) — IBM [↗](#)
- Variational Algorithm Design (2024) — IBM [↗](#)
- AI Programming with Python (2023) — Udacity [↗](#)
- Generative AI for Everyone (2023) — DeepLearning.AI [↗](#)
- XCS229 - Machine Learning (2022) — Stanford Online [↗](#)

Honors and Awards

- Two-time Recipient of Khreative U+ Creative Innovation KHU-Fellowship (2022 & 2023)
- Doctoral Research Scholarship, BrainKorea21 Plus, 2021 – 2025
- Graduate Student Scholarship, Kyung Hee University, 2021 – 2025
- National Grassroots ICT Research Initiative FYP Grant, 2020
- STOORI DA PAKHTUNKHWA Provisional Government Scholarship, 2016
- STOORI DA MARDAN District Government Scholarship, 2016

Professional References

1. Prof. Symeon Chatzinotas
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2. Prof. Hyundong Shin
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